

Benchmarking of NQCH's quantum computer

May 15, 2026

1. Report of Changes

Platform: sinq20
Calibration-id: d7f8a21692dc9acaa78c0ecabd13a51101707b4c
Calibration date: 2026-05-13 06:56:33
Calibration note: Merge pull request #17 from qiboteam/sqcal-280426
Sqcal 280426

Experiment-id: 20260514212203
Experiment date: 2026-05-14 21:22:04
Experiment note: temporary note!!!

Platform: sinq20
Calibration-id: 4dc4082f38a53222b3956c22202d32a520d4bc78
Calibration date: 2026-01-15 02:03:03
Calibration note: chore(sinq20): 2q gates cal 0-1, 0-3, 2-3, 3-4, 1-4, 4-5, 4-9, 3-8, 8-9

Experiment-id: 20260118161538
Experiment date: 2026-01-18 16:15:38
Experiment note: temporary note!!!

2. Version Comparison

Library	Version	Library	Version
qibo	0.2.23	numpy	2.4.4
qibolab	0.2.9	qibocal	0.2.5
matplotlib	3.10.8	scipy	1.17.1
scikit-learn	1.8.0	pandas	2.3.3
networkx	3.6.1	sympy	1.14.0
torch	2.11.0		

Library	Version	Library	Version
qibo	0.2.22	numpy	2.2.6
qibolab	0.2.9	qibocal	0.2.4
matplotlib	3.10.3	scipy	1.15.3
scikit-learn	1.6.1	pandas	2.2.3
networkx	3.4.2	sympy	1.14.0
torch	2.7.0		

3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



95.60

96.05

96.50

96.95

97.40

97.85

98.30

98.75

99.20

99.65

1Q Fidelity

Q17

Q18

Q19

Q12

Q13

Q14

Q15

Q16

Q7

Q8

Q9

Q10

Q11

Q2

Q3

Q4

Q5

Q6

Q0

Q1

Q17

Q18

Q19

Q12

Q13

Q14

Q15

Q16

Q7

Q8

Q9

Q10

Q11

Q2

Q3

Q4

Q5

Q6

Q0

Q1

88.65

89.85

91.05

92.25

93.45

94.65

95.85

97.05

98.25

99.45

1Q Fidelity

4. Statistics

	Average	Median	Min	Max
T1 (ns)	23.7	23.2	16.1	32
T2 (ns)	10.9	11.5	2	28.1
Fidelity	0.994	0.997	0.984	0.999
RO Fidelity	0.979	0.98	0.968	0.99
2q RB Fidelity	-	-	-	-
2q CZ Fidelity	-	-	-	-

	Average	Median	Min	Max
T1 (ns)	1.28e+04	1.23e+04	646	3.65e+04
T2 (ns)	2.36e+25	3.68e+03	125	9.43e+26
Fidelity	0.99	0.997	0.887	0.999
RO Fidelity	0.794	0.777	0.777	0.927
2q RB Fidelity	0.965	0.972	0.923	0.981
2q CZ Fidelity	0.975	0.974	0.967	0.991

5. Best Qubits Selection

k-qubits	Best Qubits	Fidelity
2	6, 11	0.939
3	6, 11, 16	0.923
4	6, 10, 11, 16	0.903
5	6, 9, 10, 11, 16	0.894

k-qubits	Best Qubits	Fidelity
2	2, 3	0.981
3	0, 2, 3	0.976
4	0, 1, 2, 3	0.970
5	0, 1, 2, 3, 4	0.965

6. Summary metrics

Benchmark	Metric	Value	Runtime
Mermin	Max	3.344	281.83 sec
Grover 2q	Max prob	0.885	3.96 sec
Grover 3q	Max prob	0.307	2.49 sec
GHZ state prep.	Success	0.902	4.66 sec

Benchmark	Metric	Value	Runtime
Mermin	Max	-3.316	1.02 sec
Grover 2q	Max prob	0.89	2.36 sec
Grover 3q	Max prob	0.385	2.43 sec
GHZ state prep.	Success	0.910	3.95 sec

7. Randomized Benchmarking Results

Qubit n	Fidelity	Error Bars	Qubit n	Fidelity	Error Bars
0	0.995	± 0.000224	10	0.996	± 0.00012
1	0.991	± 0.000662	11	0.997	± 0.000156
2	0.998	± 8.9e-05	12	0.998	± 7.26e-05
3	0.998	± 7.97e-05	13	0.998	± 9.89e-05
4	0.991	± 0.000559	14	0.998	± 5.64e-05
5	0.997	± 0.000197	15	0.998	± 0.000118
6	0.994	± 0.00018	16	0.982	± 0.00376
7	0.976	± 0.00421	17	0.956	± 0.0121
8	0.998	± 6.71e-05	18	0.995	± 0.000286
9	0.998	± 8.32e-05	19	0.996	± 0.000122

Qubit n	Fidelity	Error Bars	Qubit n	Fidelity	Error Bars
0	0.997	± 0.000131	10	0.989	± 0.000582
1	0.998	± 7.47e-05	11	0.996	± 0.000218
2	0.998	± 7.57e-05	12	0.998	± 0.000126
3	0.995	± 0.000163	13	0.998	± 7.15e-05
4	0.996	± 0.000185	14	0.985	± 0.00183
5	0.995	± 0.000147	15	0.995	± 0.000469
6	0.998	± 8.34e-05	16	0.993	± 0.000558
7	0.887	± 0.0287	17	0.99	± 0.000502
8	0.999	± 4.48e-05	18	0.997	± 0.00026
9	0.998	± 7.2e-05	19	0.998	± 9.05e-05

8. Mermin

Mermin's algorithm for 3 qubits.

Runtime	Qubits	Max
281.83 sec	16, 11, 6	3.344



Runtime	Qubits	Max
1.02 sec	0, 2, 3	-3.316



9. Grover - 2 qubits

Grover's algorithm for 2 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '11' for each pair of qubits in [[11, 6]].

Runtime	Qubits	Max prob
3.96 sec	11, 6	0.885



Runtime	Qubits	Max prob
2.36 sec	2, 3	0.89



10. Grover - 3 qubits

Grover's algorithm for 3 qubits executed on sinq20 backend with 1000 shots per circuit. We measure the success rate of finding the target state '111' for each pair of qubits in [[16, 11], [11, 6]].

Runtime	Qubits	Max prob
2.49 sec	16, 6, 11	0.307



Runtime	Qubits	Max prob
2.43 sec	0, 2, 3	0.385



11. GHZ state preparation

GHZ circuit with 3 qubits executed on sinq20 backend with 1000 shots.

Runtime	Qubits	Success
4.66 sec	16, 11, 6	0.902



Runtime	Qubits	Success
3.95 sec	0, 3, 2	0.910

